Lecture 2

Constraints and Lagrange's equations

Objectives. *Lecture 1 gave the state and the flow of a system whose particles move freely. Rods, hinges and rigid bodies constrain their motion. The allowed configurations form a* surface *in the space of particle positions. This lecture asks what Newton's law becomes on such a surface. It explains when constraint equations define a surface, identifies the allowed velocities, and states the assumption on the constraint forces. For potential forces and fixed ideal constraints, Newton's law is equivalent to* Lagrange's equations *for $\mathcal{L} = T - V$ in any coordinates on the surface. With the regularity specified below, $(q, \dot{q})$ at a given time determines the motion. The pendulum illustrates the construction.*

By the end of this lecture you will be able to:

- Determine the configuration space defined by regular constraint equations.
- Compute allowed velocities from the tangent space.
- Derive Lagrange's equations for fixed ideal constraints and potential forces.
- Formulate the pendulum dynamics in generalised coordinates.

2.1 Holonomic constraints

Constraints

From now on n is reserved for the number of *degrees of freedom*, which constraints make smaller than in Lecture 1. We write $r \in \mathbb{R}^N$ for the list of all Cartesian coordinates of all particles, so N is now the number of coordinates, three times the number of particles of Lecture 1. We write $m_i > 0$ for the mass carried by r_i (each particle's mass once per Cartesian component). A rod of length ℓ joining particles 1 and 2 forces $|\mathbf{r}_1 - \mathbf{r}_2|^2 = \ell^2$. A *holonomic constraint* is an equation on the positions alone. It does not involve the time, so we also call it *fixed*:

$$f_1(r) = \cdots = f_k(r) = 0, \qquad f = (f_1, \dots, f_k) \colon U \to \mathbb{R}^k, \tag{2.1}$$

with $U \subseteq \mathbb{R}^N$ open and f of class C^2. The *configuration space* is the set $Q = \{r \in U : f(r) = 0\}$.

Not every equation defines a surface. The set $\{xy = 0\}$ consists of two crossing lines, and $\{x^2 + y^2 = z^2\}$ is a cone. In each case the gradient vanishes at the singular point, the origin.

Definition 2.1 (Regular value)**.** Let $f \colon U \to \mathbb{R}^k$ be C^2 on an open $U \subseteq \mathbb{R}^N$. A value $c \in \mathbb{R}^k$ is *regular* if the $k \times N$ matrix $Df(r)$ has rank k at every r with $f(r) = c$; equivalently, $\nabla f_1(r), \dots, \nabla f_k(r)$ are linearly independent there. The constraints (2.1) are *regular* if 0 is a regular value.

Proposition 2.2 (Regular level sets are locally graphs). *Let c be a regular value of $f\colon U \to \mathbb{R}^k$, $Q = \{f = c\}$ and $n = N - k$. For every $r_0 \in Q$ the coordinates of $\mathbb{R}^N$ can be numbered so that $r = (y, z)$, $y \in \mathbb{R}^n$, $z \in \mathbb{R}^k$, and there are open sets $Y \subseteq \mathbb{R}^n$, $Z \subseteq \mathbb{R}^k$ with $r_0 \in Y \times Z$ and a C^2 map $g\colon Y \to Z$ with*

$$Q \cap (Y \times Z) = \{(y, g(y)) : y \in Y\}.$$

If f is C^p, so is g.

Proof. The matrix $Df(r_0)$ has rank k, so some k of its columns are independent. Call the corresponding coordinates z. Then $\partial f/\partial z(r_0)$ is invertible. The implicit function theorem says that, where this derivative is invertible, the solutions of $f(y, z) = c$ near the point form a graph $z = g(y)$, and g is C^p when f is. This gives Y, Z and g. ∎

Definition 2.3 (Surface, local parametrisation, generalised coordinates). A set $Q \subseteq \mathbb{R}^N$ is a *surface of dimension n* if every point of Q lies in the image of a *local parametrisation*: a C^2 map $r\colon Y \to \mathbb{R}^N$ on an open $Y \subseteq \mathbb{R}^n$ such that

1. r is a bijection of Y onto $Q \cap W$ for some open $W \subseteq \mathbb{R}^N$;
2. the $N \times n$ matrix $Dr(q)$ has rank n for every $q \in Y$;
3. the inverse map $q\colon Q \cap W \to Y$ is continuous.

The components of q are *generalised coordinates* on the *patch* $Q \cap W$; $n = \dim Q$ is the number of degrees of freedom.

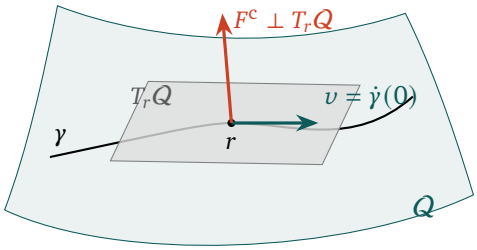


Figure 2.1: A constraint surface Q, a curve γ in it, and its tangent plane at r: the allowed velocities $T_rQ = \ker Df(r)$ (§ 2.3). An ideal constraint force (§ 2.4) is normal to it.

Corollary 2.4. *If c is a regular value of $f\colon U \to \mathbb{R}^k$, then $\{f = c\}$ is a surface of dimension $N - k$ (Figure 2.1).*

Proof. The map $r(y) = (y, g(y))$ from Proposition 2.2, with $W = Y \times Z$, has $Dr = \begin{pmatrix} \mathbb{1} \\ Dg \end{pmatrix}$ of rank n and the continuous inverse $(y, z) \mapsto y$. ∎

Regularity concerns the equations, not just the set they define. The equation $(x^2+y^2-1)^2 = 0$ defines a circle, but its gradient vanishes there. Continuity of the inverse parametrisation is needed too. The map $r(s) = (\sin 2s, \sin s)$ on $(-\pi, \pi)$ is a bijection onto a figure-eight, with $r'(s) \neq 0$, but its inverse jumps at the crossing $r(0) = (0, 0)$: points of the curve near the origin come from s near 0 and from s near $\pm\pi$ (Figure 2.2). The next lemma uses this continuity requirement.

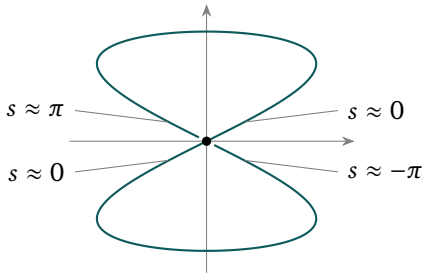


Figure 2.2: The figure-eight $r(s) = (\sin 2s, \sin s)$, $-\pi < s < \pi$. Near the crossing, one branch has s near 0 and the other has s near $\pm\pi$. So the inverse of r jumps at the crossing.

Lemma 2.5 (Changes of coordinates · *sketched*)**.** *Let $r\colon Y \to \mathbb{R}^N$ and $\tilde{r}\colon \tilde{Y} \to \mathbb{R}^N$ be local parametrisations of a surface Q, with inverses q and $\tilde{q}$. On the open set $Y_0 = r^{-1}(\tilde{r}(\tilde{Y}))$ the* transition map *$\varphi = \tilde{q} \circ r$, for which $r = \tilde{r} \circ \varphi$, is C^2 with invertible derivative.*

Proof. Near $\tilde{q}_0 = \varphi(q_0)$ some n Cartesian coordinates $\pi\colon \mathbb{R}^N \to \mathbb{R}^n$ make $D(\pi \circ \tilde{r})(\tilde{q}_0)$ invertible, so $\pi \circ \tilde{r}$ has a C^2 local inverse h (inverse function theorem). By continuity of the inverse parametrisation, $\varphi(q)$ stays in the domain of h for q near q_0, and there $\varphi = h \circ \pi \circ r$ is C^2. So is $\varphi^{-1} = q \circ \tilde{r}$; hence $D\varphi$ is invertible. ∎

2.2 Configuration spaces

The pendulum: a circle

Consider a pendulum with a rod of length ℓ in a vertical plane. Its constraint function is $f = x^2 + y^2 - \ell^2$. Since $\nabla f \neq 0$ on the circle, $Q = S^1$ and $n = 1$. The angle from the downward vertical gives the local parametrisation $r(\theta) = \ell(\sin\theta, -\cos\theta)$ on $(-\pi, \pi)$, missing the top; a second one on $(0, 2\pi)$ misses the bottom. The transition map is $\theta \mapsto \theta$ on the right half of the circle and $\theta \mapsto \theta + 2\pi$ on the left half. In practice θ is used modulo 2π (§ 1.5).

The rigid body: SO(3)

Lecture 11 studies the rigid body, and it needs this configuration space. A rigid body not contained in a line, with a point fixed at the origin, is placed by the matrix R carrying a reference placement to the current one. Two independent body directions determine R.

The matrix R preserves distances, so it is orthogonal. A motion that starts at the identity matrix $\mathbb{1}$ cannot jump to $\det R = -1$.

Here the coordinates of $\mathbb{R}^N$ are the nine entries of R, so $N = 9$, and

$$f(R) = R^{\mathsf{T}}R, \qquad Df(R)[X] = R^{\mathsf{T}}X + X^{\mathsf{T}}R,$$

where $Df(R)[X]$ is the derivative of f at R in the direction X. The values of f are symmetric matrices, so there are six equations, not nine. The value $\mathbb{1}$ is regular, since $X = \frac{1}{2}RS$ has $Df(R)[X] = S$ for symmetric S. So the orthogonal matrices form a surface of dimension $9 - 6 = 3$, and SO(3) is the part where the continuous function $\det R$ takes the value $+1$.

2.3 Velocities

Definition 2.6 (Tangent space)**.** Let Q be a surface and $r \in Q$. The *tangent space* T_rQ is the set of velocities $\dot\gamma(0)$ of C^1 curves $\gamma\colon (-\varepsilon,\varepsilon) \to \mathbb{R}^N$ with values in Q and $\gamma(0) = r$.

Theorem 2.7 (Tangent space of a regular level set)**.** *Let c be a regular value of $f\colon U \to \mathbb{R}^k$, $Q = \{f = c\}$, $r_0 \in Q$, and let $r(q)$ be a local parametrisation with $r(q_0) = r_0$. Then*

$$T_{r_0}Q = \ker Df(r_0) = \operatorname{span}\Big\{\frac{\partial r}{\partial q_1}(q_0), \dots, \frac{\partial r}{\partial q_n}(q_0)\Big\}, \tag{2.2}$$

a linear subspace of $\mathbb{R}^N$ of dimension $n = N - k$, and the vectors $\partial r/\partial q_j(q_0)$ are a basis of it.

Proof. If γ is a curve in Q through r_0, then $f(\gamma(s)) = c$ for all s, and differentiating at $s = 0$ gives $Df(r_0)\dot\gamma(0) = 0$: so $T_{r_0}Q \subseteq \ker Df(r_0)$. Conversely, for $w \in \mathbb{R}^n$ the curve $s \mapsto r(q_0 + sw)$ lies in Q and has velocity $Dr(q_0)w$: so the image of $Dr(q_0)$ lies in $T_{r_0}Q$. Hence

$$\operatorname{im} Dr(q_0) \subseteq T_{r_0}Q \subseteq \ker Df(r_0).$$

The image has dimension n because the parametrisation has rank n. The kernel has dimension $N - k = n$ by rank–nullity. So all three are equal, and the n columns of $Dr(q_0)$, which span a space of dimension n, are a basis. ∎

For the pendulum, $f = x^2 + y^2 - \ell^2$ gives $\ker Df(r) = \{v : xv_x + yv_y = 0\}$, the line spanned by $\partial r/\partial\theta = \ell(\cos\theta, \sin\theta)$. For the rigid body, $\ker Df(R) = \{X : R^{\mathsf T}X \text{ antisymmetric}\}$: the velocities at R are $\dot R = RA$ with $A^{\mathsf T} = -A$. The antisymmetric matrix A has three independent entries, which describe the body's angular velocity.

Generalised velocities and their transformation law

A motion in the patch is $r(t) = r(q(t))$, and by the chain rule

$$\dot r = \sum_{j=1}^{n} \frac{\partial r}{\partial q_j}\,\dot q_j, \tag{2.3}$$

so the *generalised velocities* $\dot q_j$ are the components of $\dot r \in T_rQ$ in the basis $\partial r/\partial q_j$. In other generalised coordinates $\tilde q = \varphi(q)$ (Lemma 2.5), $\tilde q(t) = \varphi(q(t))$ and

$$\dot{\tilde q} = D\varphi(q)\,\dot q. \tag{2.4}$$

This is the *transformation law of velocities*; Proposition 4.1 derives from it the law of momenta, which is different.

The velocity phase space

A state is a configuration with an allowed velocity: a point of

$$TQ = \{(r, v) : r \in Q,\ v \in T_rQ\} \subseteq \mathbb{R}^{2N}.$$

It is described by the equations $f(r) = 0$ and $Df(r)v = 0$. The derivative of $(r, v) \mapsto (f(r), Df(r)v)$ is block triangular, $\left(\begin{smallmatrix} Df & 0 \\ * & Df \end{smallmatrix}\right)$, of rank $2k$. So TQ is a surface of dimension $2n$, if f is C^3. Its coordinates are given by $(q, \dot q) \mapsto (r(q), Dr(q)\dot q)$. For the pendulum, TS^1 is the cylinder of § 1.5.

2.4 Ideal constraints and d'Alembert

A constraint is enforced by a force, such as the tension in a rod. Newton's law for the constrained system reads

$$m_i \ddot{r}_i = F_i + F_i^{\mathrm{c}} \qquad (i = 1, \dots, N), \tag{2.5}$$

with F the applied force and F^{c} the *constraint force*. The unknowns are n functions for the motion in Q and N for F^{c}, against N equations: without a law for F^{c} the motion is not determined.

Definition 2.8 (Virtual displacement, ideal constraint)**.** A *virtual displacement* at $r \in Q$ is a vector $\delta r \in T_r Q$: a displacement the constraint allows. The constraint is *ideal* if the constraint force does no virtual work:

$$F^{\mathrm{c}} \cdot \delta r = 0 \qquad \text{for every } \delta r \in T_r Q. \tag{2.6}$$

Together, (2.5) and (2.6) say $\sum_i (m_i \ddot{r}_i - F_i)\,\delta r_i = 0$ for every virtual displacement: *d'Alembert's principle*.

This is an axiom about the constraint, not a theorem, and it is what "frictionless" means. By Theorem 2.7 it says F^{c} is orthogonal to $\ker Df$. The orthogonal complement of $\ker Df$ is the span of the rows of Df, by the linear-algebra identity $(\ker A)^{\perp} = \operatorname{im} A^{\mathsf{T}}$. Thus

$$F^{\mathrm{c}} = \sum_{\alpha=1}^{k} \lambda_\alpha \nabla f_\alpha \tag{2.7}$$

for some numbers λ_α, the *Lagrange multipliers*: an ideal constraint force is normal to Q (Figure 2.1). For a rod joining particles 1 and 2, $\nabla |\mathbf{r}_1 - \mathbf{r}_2|^2 = \big(2(\mathbf{r}_1 - \mathbf{r}_2),\, 2(\mathbf{r}_2 - \mathbf{r}_1)\big)$: the forces on the two ends are equal, opposite and along the rod. Friction is not of this form.

Along an actual motion $\dot r \in T_r Q$, so an ideal constraint force does no work on the motion either: $F^{\mathrm{c}} \cdot \dot r = 0$.

2.5 Lagrange's equations, and the pendulum

Let the applied force be $F = -\nabla V(r,t)$, the constraints $f(r) = 0$ be C^2 and regular, and $r(q)$ a local parametrisation of Q (Proposition 2.2 supplies one near each point). Call $v(q,\dot q) = Dr(q)\,\dot q$ the right-hand side of (2.3), with $\dot q$ an independent variable, and define the *Lagrangian*

$$T(q,\dot q) = \tfrac{1}{2} \sum_{i=1}^{N} m_i\, v_i(q,\dot q)^2, \qquad V(q,t) = V\big(r(q),t\big), \qquad \mathcal{L} = T - V. \tag{2.8}$$

Lemma 2.9 (Two identities)**.** *With v as above,*

$$(a)\quad \frac{\partial v}{\partial \dot q_j} = \frac{\partial r}{\partial q_j}, \qquad (b)\quad \frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{\partial r}{\partial q_j}\big(q(t)\big)\right] = \frac{\partial v}{\partial q_j}\big(q(t),\dot q(t)\big)$$

for every C^2 curve $q(t)$.

Proof. (a): v is linear in $\dot q$, with coefficient $\partial r/\partial q_j$ on $\dot q_j$. (b): both sides equal $\sum_l \frac{\partial^2 r}{\partial q_l \partial q_j}\dot q_l$, the left by the chain rule, the right by differentiating v in q_j; mixed partial derivatives of a C^2 map commute. ∎

Theorem 2.10 (Lagrange's equations)**.** *Under the hypotheses above, let $q(t)$ be a C^2 curve in the parameter domain and $r(t) = r(q(t))$. The following are equivalent:*

1. (Newton with an ideal constraint.) $m_i \ddot{r}_i = -\partial V/\partial r_i + F_i^{\mathrm{c}}$ *with* $F^{\mathrm{c}}(t) \perp T_{r(t)}Q$ *for all* t*;*
2. (Lagrange.) *for* $j = 1, \dots, n$*,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \frac{\partial \mathcal{L}}{\partial q_j} = 0. \tag{2.9}$$

In either case the constraint force is determined by the motion: $F^{\mathrm{c}} = m\ddot{r} + \nabla V$, componentwise.

Proof. For any C^2 curve, let P_j be the component of the residual $m\ddot{r} + \nabla V$ along the jth basis vector:

$$P_j = \sum_i \Big(m_i \ddot{r}_i + \frac{\partial V}{\partial r_i} \Big) \frac{\partial r_i}{\partial q_j}.$$

Since $\dot{r} = v$, the two identities of Lemma 2.9 give

$$\sum_i m_i \ddot{r}_i \frac{\partial r_i}{\partial q_j} = \frac{\mathrm{d}}{\mathrm{d}t}\Big(\sum_i m_i v_i \frac{\partial v_i}{\partial \dot{q}_j} \Big) - \sum_i m_i v_i \frac{\partial v_i}{\partial q_j} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j}.$$

By the chain rule $\sum_i \frac{\partial V}{\partial r_i}\frac{\partial r_i}{\partial q_j} = \frac{\partial V}{\partial q_j}$, and V does not depend on $\dot{q}$. Hence

$$P_j = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \frac{\partial \mathcal{L}}{\partial q_j}.$$

(1)⇒(2): the residual is F^{c}, which is orthogonal to each $\partial r/\partial q_j \in T_r Q$; so $P_j = 0$. (2)⇒(1): define $F^{\mathrm{c}} = m\ddot{r} + \nabla V$. Then $F^{\mathrm{c}} \cdot \partial r/\partial q_j = P_j = 0$ for all j, and by Theorem 2.7 the $\partial r/\partial q_j$ span $T_r Q$; so $F^{\mathrm{c}} \perp T_r Q$. ∎

Thus Lagrange's equations are obtained by taking the scalar product of Newton's law with each basis vector $\partial r/\partial q_j$. The constraint force disappears because it is orthogonal to those vectors. The theorem holds in *every* patch. On an overlap, both sets of equations describe the same motions because both are equivalent to Newton's law with an ideal constraint force. Lecture 3 derives this invariance directly from Lagrange's equations.

Corollary 2.11 (Determinacy on the constraint surface · *proved modulo T1 (cited)*)**.** *Write $M = Dr^{\mathsf{T}} \operatorname{diag}(m_i)\, Dr$, an $n \times n$ matrix, where $Dr = \partial r/\partial q$. Then M is positive definite, and (2.9) can be solved for $\ddot{q}$:*

$$\ddot{q} = M(q)^{-1}\, \Phi(q, \dot{q}, t)$$

for a function Φ built from r, V and their derivatives. If r and V are C^3, this is a C^1 vector field on the coordinates $(q, \dot{q})$ (which fix the velocity through (2.3)), and T1 of Lecture 1 (Theorem 1.3: unique maximal solutions) applies: $(q, \dot{q})$ at a time t_0 determines the motion for as long as it stays in the patch, and with it the constraint force.

Proof. We have $\dot{q}^{\mathsf{T}} M \dot{q} = \sum_i m_i (Dr\,\dot{q})_i^2 > 0$ for $\dot{q} \neq 0$, as Dr has rank n. The kinetic energy T is quadratic in $\dot{q}$ with Hessian M, so $\partial \mathcal{L}/\partial \dot{q} = M\dot{q}$ and (2.9) reads $M\ddot{q} = \Phi$. The function Φ contains second derivatives of r, C^1 when r is C^3; the parametrisations of Proposition 2.2 are C^3 when f is. If V depends on t, the field is made autonomous as in (1.6). ∎

The axiom of ideal constraints restores determinacy: for the C^3 data of Corollary 2.11, the pair $(q, \dot{q})$ at a time t_0 determines a unique maximal motion in the patch, and the constraint force follows from it.

Example 2.12 (No constraint: curvilinear coordinates)**.** With $k = 0$ the theorem is Newton's law in arbitrary coordinates. A free particle in the plane in polar coordinates, $\rho > 0$, has $\mathcal{L} = \frac{1}{2}m(\dot\rho^2 + \rho^2\dot\varphi^2)$, and (2.9) gives

$$m\ddot\rho = m\rho\dot\varphi^2, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\big(m\rho^2\dot\varphi\big) = 0,$$

valid until the path reaches the origin. The centrifugal term comes from the use of curvilinear coordinates, not an additional applied force.

The pendulum

To apply the theorem, choose coordinates and write the parametrisation $r(q)$. Express the kinetic energy T and the potential energy V in $(q, \dot q)$, then apply (2.9).

Take $r = \ell(\sin\theta, -\cos\theta)$, with gravity $-mg$ in the second coordinate y. The kinetic and potential energies are

$$T = \frac{1}{2}m\ell^2\dot\theta^2, \qquad V = mgy = -mg\ell\cos\theta.$$

Hence the Lagrangian and Lagrange's equation are

$$\mathcal{L} = \frac{1}{2}m\ell^2\dot\theta^2 + mg\ell\cos\theta, \qquad m\ell^2\ddot\theta + mg\ell\sin\theta = 0.$$

In units with $m = g = \ell = 1$ this is $\ddot x = -\sin x$ of (1.12).

The rod's tension, which the equation never mentions, follows from the motion, as the theorem says. Differentiating r twice and using the equation of motion in $F^{\mathrm{c}} = m\ddot r + \nabla V$ gives

$$F^{\mathrm{c}} = \big(m\ell\dot\theta^2 + mg\cos\theta\big)(-\sin\theta, \cos\theta).$$

The vector $(-\sin\theta, \cos\theta)$ is the unit vector along the rod towards the pivot. So the constraint force is normal to the circle, as (2.7) requires. Its coefficient $m\ell\dot\theta^2 + mg\cos\theta$ is the tension: positive when the rod pulls towards the pivot, and negative when the rigid rod pushes away from it.

Summary.

- *Handed on:* the configuration space Q, a surface of dimension n covered by patches of generalised coordinates with smooth transitions (Corollary 2.4 and Lemma 2.5); its velocities $T_rQ = \ker Df$, which transform by $D\varphi$ (Theorem 2.7, (2.4)).
- *The main result:* for fixed regular ideal constraints and potential forces, Newton's law is equivalent to Lagrange's equations for $\mathcal{L} = T - V$, in any patch (Theorem 2.10).
- *Determinacy:* for C^3 data, $(q, \dot q)$ at a time t_0 determines the motion for as long as it stays in the patch (Corollary 2.11).
- *Next:* Lecture 3 asks why the *form* of these equations survives every change of coordinates: they make one number, the action, stationary.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §§17–18 (constraints and examples, pp. 75–83) and §21 (d'Alembert, pp. 91–97). Tong, *Lectures on Classical Dynamics*, §2.3 (constraints, pp. 17–21).